

Mohammed Alyafeai

Applied AI Researcher & Engineer

 Alyaf3i1118@gmail.com

 0503259118

 <https://www.linkedin.com/in/mohammed-alyafeai/>

 <https://Alyaf3i.github.io/>

Summary

AI practitioner working on large language models and agentic systems, combining model-level iteration with practical deployment. Experience includes fine-tuning, evaluation, and shaping model behavior, alongside building tools and pipelines that enable agents to operate reliably in real-world settings. Work also explores challenges specific to Arabic language use.

Experience

Senior Associate AI Engineer [TII – Abu Dhabi, UAE]

Dec 2023 – Present

- Trained **Falcon-Arabic & H1-Arabic**, both which were a state-of-the-art Arabic LLMs at the time (top in their size category).
- Led three cross-functional AI initiatives end-to-end, defining modeling approaches, evaluation methodology, and system architecture, while aligning research experimentation with production delivery.
- Unified research and engineering by designing scalable data generation, RAG, and web-search pipelines, establishing consistent experimentation and evaluation practices across projects.
- Led the design and development of a tool-calling system, implementing multi-step reasoning workflows (query rewriting, retrieval, and tool orchestration) to enable reliable task execution.
- Designed and evaluated multiple retrieval and modeling approaches through structured experimentation (including ablations and benchmarking), using quantitative results to guide architecture decisions in client-facing POCs.
- Developed agentic pipelines involving multi-step reasoning (query rewriting, retrieval, filtering, and response generation), improving system accuracy and robustness across real-world use cases.
- Co-led development of Falcon-Arabic (state-of-the-art Arabic LLM), contributing to model architecture decisions, evaluation strategy, and production-readiness considerations.
- Acted as technical manager for a production chat application, managing outsourced engineering teams through weekly alignment meetings and ensuring delivery met architectural and product requirements.
- Mentored new engineers and researchers, establishing onboarding guidance, technical standards, and review processes.
- Built and standardized LLM evaluation frameworks, defining task-specific metrics and custom scoring methods to better capture real-world performance.
- Designed and implemented scalable cloud and serverless workflows supporting MLOps and automated experimentation pipelines.
- Optimized cloud resource allocation across high-cost compute nodes, introducing automation that reduced infrastructure waste and saved over \$5K per day.
- Presented structured technical progress updates and strategic plans to senior leadership, ensuring alignment between AI research, engineering execution, and organizational priorities.

AI/ML & Backend Engineer [Aixplain – California, US] [remotely]

Feb 2023 – Feb 2024

- Integrated multiple ML models using KServe, exposing them through user-friendly APIs for inference and onboarding.
- Containerized model pipelines to support synchronous and asynchronous execution.
- Configured and deployed services on Kubernetes, supporting scalable model hosting and inference workflows.
- Utilized AWS ECR for container images and AWS S3 for model assets and artifacts.
- Built backend systems leveraging vector databases (Qdrant) for similarity search and retrieval use cases.
- Implemented end-to-end CI/CD pipelines using GitHub Actions and custom scripts to automate solution onboarding.
- Took ownership of a GitHub repository, contributing the majority of features and performing code reviews.
- Conducted extensive SDK testing, fixed bugs, and automated documentation generation using Sphinx.

Data Analyst & Software Engineering [Cyber Gate Defense – Abu Dhabi, UAE]

Sep 2021 - Mar 2023

- Built and automated data ingestion pipelines from multiple SIEM solutions.
- Developed ML, deep learning, and NLP models for security and analytics use cases.
- Automated generation of Power BI reports using Python.
- Developed and maintained a data visualization portal for monitoring customer environments.
- Managed Elasticsearch clusters, indexed data, and visualized insights using Kibana.
- Parsed logs, JSON, and YAML data using Logstash, and implemented watchers for device monitoring.
- Documented technical specifications and maintained clean, reproducible codebases.

Education

Al Ain University
Bachelor of Software Engineering 2018-2021 (honored)

Khalifa University
Master in Computer Science 2025-2026

Projects & Contributions

- **Falcon-Arabic:** State-of-the-art Arabic LLM in its size category, co-developed as part of a two-person team.
- **Tool-Calling System:** Led the development of a tool-calling system with a robust pipeline, achieving measurable improvements over baseline approaches in internal evaluations.
- **Private Translation Model:** Developed a private state-of-the-art translation model specialized in Arabic translation.
- **Document Search Engine by Using NLP:** Search Engine that search for an answer from set of documents. [Python, NLP, fine-tuning, Django]
- **IDRF:** A program that combines the AI and GIS to find the best path in the desert. [Python, AI, Graphs, Shortest Path, Sorting]
- **Model for diacritization:** Manage to build and tune a model that diacritize Arabic text depending in UAE dialect. Worked in fine tuning it to get a good accuracy that qualified me to win the first place in competition of Sibaq Lahja. [Python, NLP by using Deep learning]
- **AutoAuditor:** A web app that use LLMs to do security auditing. [LLMs, Python, Django, redis, celery]
- **Chatbot Generator:** A Website which enable user to create his own chatbot and provide the user with API.[Python, vector database, NLP]
- **Agentic RAG:** Designed a fully open-source agentic RAG assignment architecture with OpenWebUI frontend, Python backend APIs, document indexing, evaluation, observability, and Docker Compose deployment; baseline stack includes Docling, LlamaIndex, PGVector/PostgreSQL, Ollama, CrewAI, Arize Phoenix, and RAGAs.

Skills

Key Competencies:

- Leading and mentoring
- Taking initiative
- System Design
- Teamwork

Languages:

- Arabic (Native)
- English (fluent)

AI & Data Analysis Skills:

- Hands-on experience across classical ML, deep learning, and NLP, including classification, retrieval, translation, conversational AI, and information retrieval systems.
- Hands-on experience building machine learning and deep learning models using scikit-learn, and PyTorch.
- Experience developing large language models (LLMs) and designing robust tools and workflows to enable their use in production.
- Deep learning fundamentals, including implementing models from scratch using NumPy for mathematical understanding.

Programming languages & Frameworks:

- Python
- Elasticsearch
- Django
- Pytest
- C
- MongoDB
- Flask
- Kubernetes
- S3
- Docker
- GCP
- regex

Publications

Research focus: Large language models for Arabic, with emphasis on evaluation, benchmarking, and model adaptation.

- **Alyafeai, M.,** Alobeidli, H., Alkaabi, O., Alsuwaidi, S., Al Qadi, L., Alzubaidi, A., Alhammadi, M., Boussaha, B. E. A., Hacid, H. **Hamsa: A Manually Annotated Emirati Arabic Corpus for Speech and Language Technologies.** Proceedings of INTERSPEECH 2026, Sydney, Australia, 2026.
- Boussaha, B. E. A., **Alyafeai, M.,** Alzubaidi, A., Al Qadi, L., Alsuwaidi, S., Hacid, H. *Adapting Falcon3-7B Language Model for Arabic: Methods, Challenges, and Outcomes.* Proceedings of the **Third Arabic Natural Language Processing Conference (ArabicNLP)**, Suzhou, China, 2025, pp. 1–15.
- Malartic, Q., Roy Chowdhury, N., Cojocaru, R., Farooq, M., Campesan, G., Djilali, Y. A. D., Narayan, S., Singh, A., Velikanov, M., Boussaha, B. E. A., **Al-Yafeai, M.,** Alobeidli, H., Al Qadi, L., et al. *Falcon2-11B Technical Report.* arXiv preprint, 2024. arXiv:2407.14885

- Boussaha, B. E. A., Al Qadi, L., Farooq, M., Alsuwaidi, S., Campesan, G., Alzubaidi, A., **Alyafeai, M.**, Hacid, H. *3LM: Bridging Arabic, STEM, and Code through Benchmarking*. Proceedings of the **Third Arabic Natural Language Processing Conference (ArabicNLP)**, Suzhou, China, 2025, pp. 42–63.
- Alhelli, L., **Al-Yafeai, M.**, Daoud, M. S., Elamsy, T. *An Intelligent Route Finder for UAE Desert Drivers Based on the A* Algorithm*. Proceedings of the **8th International Conference on IoT: Systems, Management and Security (IOTSMS)**, Gandia, Spain, 2021, pp. 1–5. <https://doi.org/10.1109/IOTSMS53705.2021.9704931>
- Alabbasi, N., Al-Badrashiny, M., Aldahmani, M., AlDhanhani, A., Alhashmi, A. S., Alhashmi, F. A., Al Hashemi, K., Alkhobbi, R. E., Al Maazmi, S. T., **Alyafeai, M. A.**, Alzaabi, M. M., Alzaabi, M. S., et al. *Gulf Arabic Diacritization: Guidelines, Initial Dataset, and Results*. Proceedings of the **Seventh Arabic Natural Language Processing Workshop (WANLP)**, Abu Dhabi, 2022. <https://doi.org/10.18653/v1/2022.wanlp-1.33>
- Daoud, M. S., Abu Shawar, B., **Alyafeai, M.**, Essam, A., Ghassan, M., Alzaabi, N. *Enhancing Intrusion Detection Systems Accuracy Using Machine Learning*. Proceedings of the **International Conference on Software Defined Systems (SDS)**, 2023, pp. 103–106. <https://doi.org/10.1109/SDS59856.2023.10329276>
- *Are Arabic Benchmarks Reliable? QIMMA's Quality-First Approach to LLM Evaluation*. [submitted]
- *Evaluating Arabic Large Language Models: A Survey of Benchmarks, Methods, and Gaps*. [submitted]

Certifications & Achievements

- **Developed Internal SOTA English-to-Arabic Translation Model Based on Falcon-H1 Arabic (Feb 2026)**
 - Owned the end-to-end fine-tuning pipeline of a private English-to-Arabic translation model on top of Falcon-H1 Arabic, achieving state-of-the-art performance compared to models of similar size and models up to x3 larger.
- **Release Falcon H1 Arabic (Jan 2026)**
 - Primary contributor after the lead, responsible for the end-to-end continuous pre-training of Falcon-H1 Arabic.
- **Release Falcon Arabic (May 2025)**
 - Core contributor responsible for data preparation, tokenization strategy, and evaluation-driven refinement, systematically improving model robustness and performance..
- **Won Sibaq Lahja Program (Oct 2022)**
 - Participated In Sibaq Lahja, and manage to win the first for building a model that diacritize gulf Arabic text.
- **Managing Teams – Project Management Institute (Apr 2025)**
 - Focused on team leadership, delegation strategy, performance alignment, and organizational effectiveness.
- **Managing Team Conflict – Project Management Institute (Apr 2025)**
 - Training in structured conflict resolution, stakeholder alignment, and maintaining high-performing engineering environments.
- **Enhancing Team Innovation – Project Management Institute (Apr 2025)**
 - Focused on fostering innovation culture, structured ideation, and scalable team creativity practices.
- **Strategic Communication for Leaders (Mar 2025)**
 - Advanced leadership communication, executive reporting, and cross-functional influence strategies.
- **Project Management Integration & Quality (Sep–Oct 2024)**
 - Covered governance frameworks, risk management, integration strategy, and quality assurance at the organizational level.